

Cletus Takudzwa Ngwerume

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Professional Summary

Data scientist and software engineer with 5+ years of experience designing and operationalizing machine learning and distributed systems. Expertise in Python, PyTorch, scikit-learn, SQL and cloud deployments on AWS using Docker and Kubernetes. Proven track record building enterprise platforms, fine-tuning LLMs and deploying reproducible ML pipelines to production.

Education

Michigan Technological University

Master of Science, Data Science

- **GPA:** 4.0

Apr 2026

Houghton, MI

Harare Institute of Technology

Bachelor of Technology, Computer Science

- **Achievements:** First Class

Harare, Zimbabwe

Technical Skills

- **Programming Languages:** Python, R, SQL, JavaScript, C#, Visual Basic
- **Machine Learning, AI & Data Analysis:** Classification, Clustering, Feature Engineering, Model Validation, ROC/AUC, Confusion Matrices, Deep Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Data Analysis, Generative AI, Model Lifecycle Management
- **Frameworks & Libraries:** PyTorch, Hugging Face Transformers, Scikit-learn
- **Distributed Systems & Data:** REST APIs, Microservices, Data Pipelines, Vector Databases (pgvector)
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Cloud & DevOps:** AWS, Docker, CI/CD, Nginx, Supervisor
- **Development Practices & Core Competencies:** Object-Oriented Design, Design Patterns, Code Reviews, Agile/Scrum, SDLC, Curiosity, Life Insurance Experience
- **Operations:** Monitoring, Incident Response, Root-Cause Analysis, Production Support

Professional Experience

Michigan Technological University

Research Assistant - Large Language Models

Houghton, MI • **May 2025 - Present**

- Fine-tune LLMs with PyTorch on AWS instances across 3B-70B parameters, manage end-to-end model lifecycle from training to deployment, benchmark accuracy (ROC/AUC), latency, and resource utilization to identify the model size that meets performance targets
- Designed reproducible experiment pipelines for large-scale model comparison and statistical evaluation using data analysis techniques
- Conducted ablation studies on chunking strategies, embedding models, and retrieval configurations to support model validation

Research Assistant - Human Robot Interaction

Houghton, MI • **Aug 2024 - Oct 2025**

- Integrated LLM-driven natural language understanding using Hugging Face Transformers and Python into the robot's control software, enabling speech-based interaction that let users issue commands verbally, which allowed the robot to respond without manual input
- Built and evaluated speech-to-text (STT) and text-to-speech (TTS) pipelines for real-time robotic responses
- Collaborated with interdisciplinary teams to align AI behavior with human-centered design principles, demonstrating curiosity

LADS Africa

Provincial Project Manager / Technical Lead

Feb 2023 - Aug 2024

- Led the end-to-end development of a custom enterprise ERP platform using Python, C#, Docker, and design patterns, coordinating requirements gathering, system architecture, testing, and deployment, which delivered a stable solution on schedule and reduced post-launch defects
- Replaced SAP as the organization's core system, reducing enterprise software and licensing costs by 60%
- Designed scalable, service-oriented architectures and data pipelines supporting finance, billing, analytics, and reporting workflows
- Established CI/CD pipelines using Docker and Kubernetes, enforced code reviews, and produced technical documentation to ensure maintainability
- Monitored production systems, diagnosed incidents, and drove root-cause resolution to maintain operational stability
- Collaborated closely with non-technical stakeholders to translate business needs into scalable technical solutions

Intelli Africa Solutions

Head of Software Development

May 2021 - Feb 2023

- Architected and delivered distributed microservices and backend systems using Python and Apache Spark, enabling digital financial services across 200+ remote locations
- Increased platform reach by 20%+ by designing containerized systems resilient to low-bandwidth and intermittent connectivity

- Designed data models, APIs, and synchronization mechanisms using object-oriented design and clustering techniques, which improved reliability of distributed data management
- Led agile development cycles including sprint planning, implementation, testing, deployment, and post-release support
- Implemented monitoring and logging to ensure availability across geographically distributed deployments

LADS Africa

Senior Software Engineer

Mar 2020 - May 2021

- Designed and implemented scalable municipal billing and revenue collection systems used by 1M+ users
- Reduced operational costs by 40% and increased SME revenue collection by 60% through automation and data pipelines
- Developed and optimized SQL database schemas and queries for high-volume transactional workloads, improving query performance and supporting faster data access
- Built secure, fault-tolerant integrations with external banking and payment systems using REST APIs and Docker containers
- Participated in on-call rotations, production troubleshooting, and incident response

Selected Projects

LLM-Powered Question Answering and RAG Platform

- Designed and built an end-to-end AI system covering data ingestion, preprocessing, embedding, retrieval, inference, and evaluation
- Implemented modular components to support experimentation with multiple models and retrieval strategies
- Evaluated accuracy, latency, and cost trade-offs to guide deployment decisions
- Followed clean code practices, documentation standards, and reproducible workflows

COVID-19 ICU Admission Risk Prediction

- Built and validated classification models using structured clinical datasets
- Performed feature engineering, cross-validation, and threshold tuning
- Interpreted confusion matrices and ROC curves to assess model readiness